

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| <b>Ex.No.7</b> | <b>Use AF Logical OSE to extract data from an Android device</b> |
| <b>Date</b>    |                                                                  |

## Description

AF Logical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AF Logical OSE to extract data from an Android device.

### Step 1: Prepare Your Environment

#### 1. Download AF Logical OSE:

- Visit the [AF Logical OSE GitHub page](#) and download the latest release. Alternatively, you can clone the repository using Git.

#### 2. Install Java (if not already installed):

- AF Logical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).

#### 3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the [Android Developer's website](#) and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

#### 4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- Go to Settings > Developer Options and enable USB Debugging.

### Step 2: Connect the Android Device to Your Computer

#### 1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

#### 2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:  
Bash

adb devices

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

### **Step 3: Extract Data Using AF Logical OSE**

#### **1. Run AF Logical OSE:**

- Navigate to the directory where you have downloaded or extracted AF Logical OSE.

#### **2. Push the APK to the Android Device:**

- Run the following command to install the AF Logical OSE APK on the device:  
Bash  
adb install af\_logical.apk
- This installs the AFLogical app on the device, which will be used to collect data.

#### **1. Launch the AF Logical OSE App on the Device:**

- On the Android device, open the AF Logical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

#### **2. Start Data Extraction:**

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in .csv files on the device's storage, typically in a directory named af\_logical.

### **Step 4: Transfer Extracted Data to Your Computer**

#### **1. Use ADB to Pull Data:**

- After extraction, you need to transfer the data files from the Android device to your computer. Run:  
bash  
adb pull /sdcard/aflogical /path/to/destination
- Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.

#### **1. Verify the Data:**

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

### **Step 5: Analyze the Extracted Data**

## 1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files.
- These files contain the extracted data like contacts, SMS, MMS, and call logs.

## 2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

## Step 6: Clean Up

### 1. Uninstall AF Logical OSE:

- Once data extraction is complete, you can uninstall the AF Logical OSE app from the Android device by running:

```
bash
```

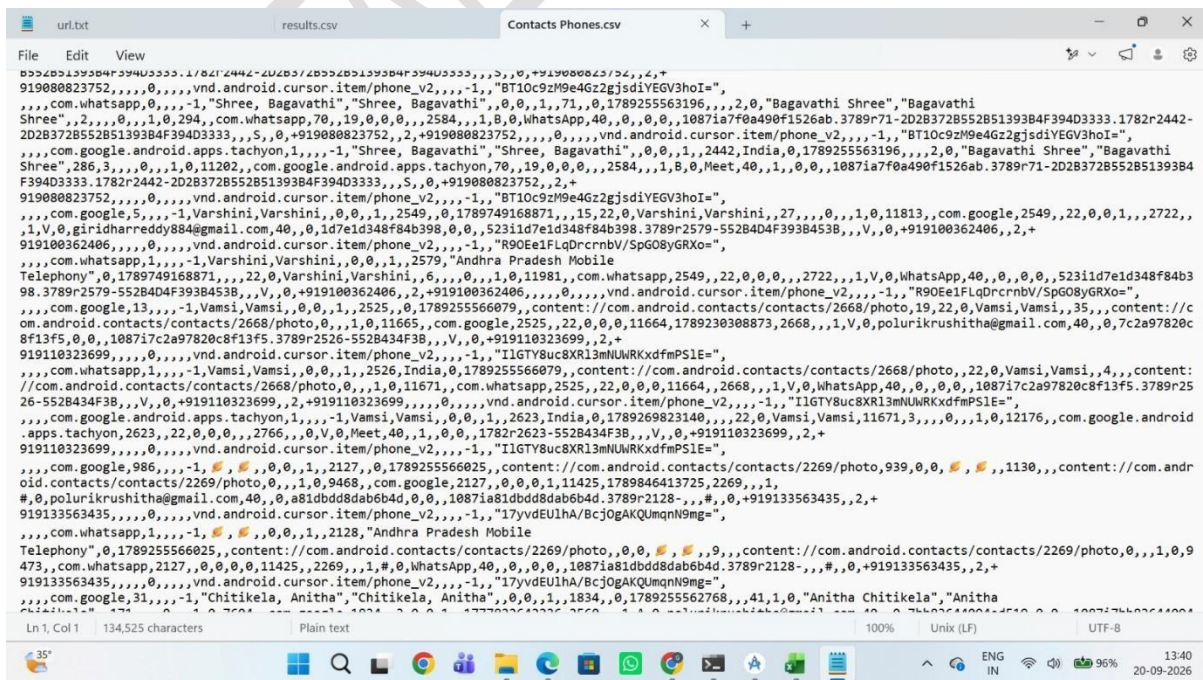
```
adb uninstall com.viaforensics.android.aflogical
```

### 2. Disconnect the Device:

- Safely disconnect the Android device from your computer.

By following these steps, you can use AF Logical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

## Output Screenshots:



```

C:\Users\user\Downloads\platform-tools-latest-windows>cd platform-tools

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb devices
List of devices attached
10BE6X06HH000ED device

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb install -r "C:\Users\user\AndroidStudioProjects\AFLogicalOSE
2\app\build\outputs\apk\debug\app-debug.apk"
Performing Streamed Install
Success

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb shell ls /sdcard/Android/data/com.viaforensics.android.aflog
ical_ose/files/forensics
20260920.1240
20260920.1242

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb shell ls -R /sdcard/Android/data/com.viaforensics.android.af
logical_ose/files/forensics
/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics:
20260920.1240
20260920.1242

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1240:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1242:
info.xml

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb logcat -c

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb logcat -d -t 500 | findstr /I "AndroidRuntime FATAL EXCEPTIO
N SecurityException IllegalArgumentException viaforensics aflogical"

```

```

C:\Windows\System32\cmd X + v
logical_ose/files/forensics
/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics:
20260920.1240
20260920.1242
20260920.1248
20260920.1300
20260920.1332

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1240:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1242:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1248:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1300:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1332:
Contacts Phones.csv
info.xml

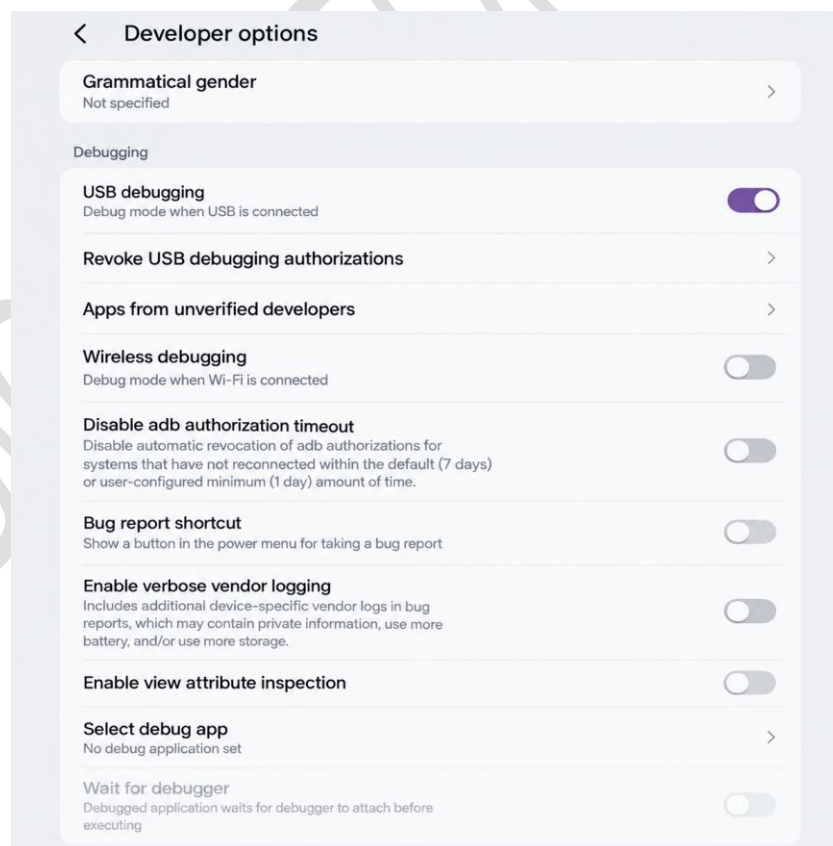
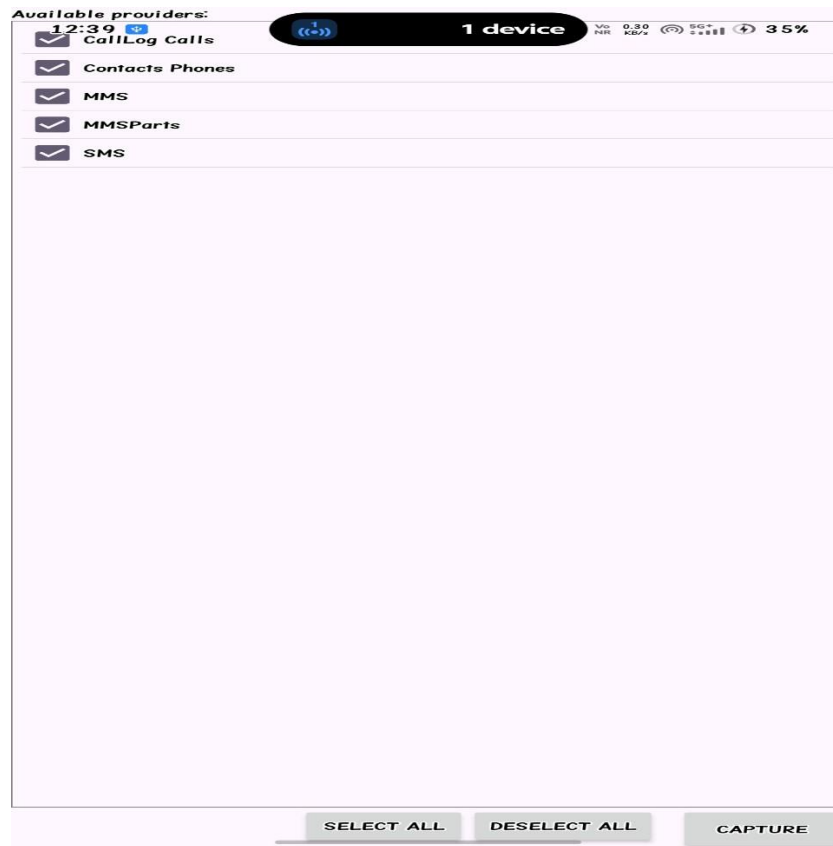
C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>if not exist C:\AFLogical_Result mkdir C:\AFLogical_Result

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb pull /sdcard/Android/data/com.viaforensics.android.aflogical
_ose/files/forensics/ C:\AFLogical_Result
/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files...cs/: 6 files pulled, 0 skipped. 5.8 MB/s (288084 bytes in 0.047s)

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>explorer C:\AFLogical_Result

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>

```



**Rubrics:**

| Criteria & Marks Assigned                             | Mark Allotted | Mark Awarded |
|-------------------------------------------------------|---------------|--------------|
| 1.GitHub Activity & Submission Regularity             | 3             |              |
| 2.Application of Forensic Tools & Practical Execution | 3             |              |
| 3. Documentation & Reporting                          | 2             |              |
| 4. Engagement, Problem-Solving & Team Collaboration   | 2             |              |
| Total                                                 | 10            |              |

**Result:**

AF Logical OSE was used to perform logical data extraction from an Android device. The extracted data was transferred to the computer and analyzed in CSV format for forensic investigation.